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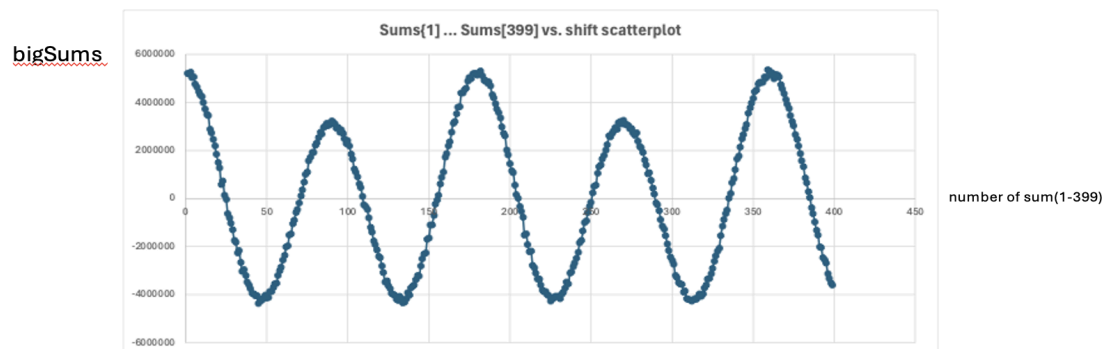
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Project Number: #7

Project Name: Autocorrelation using MPI

Project Report #7

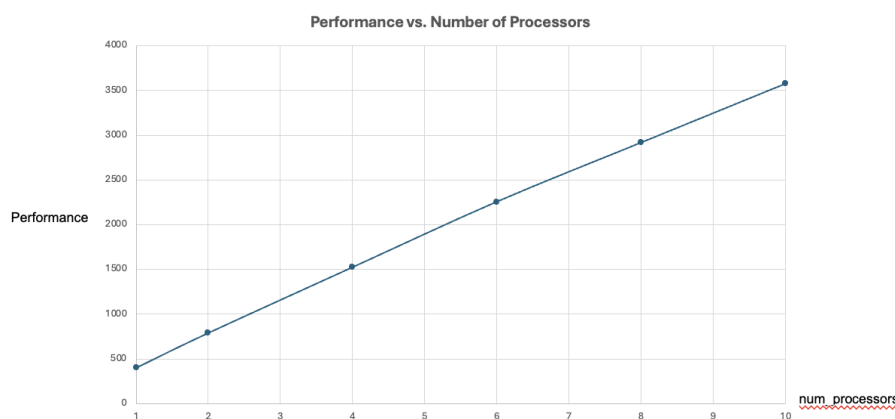
1. Show the Sums[1] ... Sums[399] vs. shift scatterplot.



2. State what the two secret sine-wave periods are.

The first secret sine-wave period is 178 and the second secret sine-wave period is 91.

3. Show your graph of Performance vs. Number of Processors **used**.



4. What patterns are you seeing in the performance graph?

The pattern I found from the graph above is that as the number of processors increases, the performance also increases.

5. Why do you think the performances work this way?

The performance increases with the number of processors because parallel computing allows tasks to be divided among multiple processors, reducing the workload on each individual processor. This division of labor leads to faster computation times and improved overall efficiency. By distributing the data and computational tasks, the system can handle larger datasets and more complex calculations more effectively. Additionally, each processor can work simultaneously on different parts of the problem, which leads to significant reductions in total computation time.